Homework 3

Branda Sisoutham

2026-09-15

Question 2

```
head(gala)
```

```
##              Species Endemics  Area Elevation Nearest Scruz Adjacent
## Baltra            58       23 25.09       346     0.6   0.6     1.84
## Bartolome         31       21  1.24       109     0.6  26.3   572.33
## Caldwell           3        3  0.21       114     2.8  58.7     0.78
## Champion          25        9  0.10        46     1.9  47.4     0.18
## Coamano            2        1  0.05        77     1.9   1.9   903.82
## Daphne.Major      18       11  0.34       119     8.0   8.0     1.84
```

```
gala_glm = glm(Species ~ Area + Elevation + Nearest + Scruz + Adjacent,
               family = poisson,
               data = gala)

summary(gala_glm)
```

```
##
## Call:
## glm(formula = Species ~ Area + Elevation + Nearest + Scruz +
##     Adjacent, family = poisson, data = gala)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  3.155e+00  5.175e-02  60.963  < 2e-16 ***
## Area        -5.799e-04  2.627e-05 -22.074  < 2e-16 ***
## Elevation    3.541e-03  8.741e-05  40.507  < 2e-16 ***
## Nearest      8.826e-03  1.821e-03   4.846 1.26e-06 ***
## Scruz       -5.709e-03  6.256e-04  -9.126  < 2e-16 ***
## Adjacent    -6.630e-04  2.933e-05 -22.608  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 3510.73  on 29  degrees of freedom
## Residual deviance:  716.85  on 24  degrees of freedom
## AIC: 889.68
##
## Number of Fisher Scoring iterations: 5
```

The model gives us 3.15 as the intercept, -0.0005799 for Area, 0.003541 for Elevation, 0.008826 for Nearest, -0.005709 for Scruz, and -0.0006630 for Adjacent. The residual deviance was 716.85.

$$\eta = \log(\mu)$$

$$\frac{d\eta}{d\mu} = \frac{1}{\mu}$$

The variance function is

$$V(\mu) = \mu$$

The weights are

$$w = \mu$$

The adjusted dependent variable is

$$z = \log(\mu) + \frac{y - \mu}{\mu}$$

```
y = gala$Species
mu = (y + mean(y)) / 2

w = mu
z = log(mu) + (y - mu) / mu
```

```
fit1 = lm(z ~ Area + Elevation + Nearest + Scruz + Adjacent,
          weights = w, data = gala)

coef(fit1)
```

```
##   (Intercept)          Area     Elevation       Nearest         Scruz
##  3.2943733417 -0.0005501185  0.0034563391  0.0032966580 -0.0029947394
##      Adjacent
## -0.0007104086
```

In the first iteration, the estimated coefficients are 3.29437, -0.0005501, 0.0034563, 0.0032967, -0.0029947, and -0.0007104. These are close to the coefficients from the GLM fit, but there are some differences.

```
eta1 = fitted(fit1)
mu1 = exp(eta1)
dev1 = 2 * sum(y * log(y / mu1) - (y - mu1))
dev1
```

```
## [1] 802.1661
```

After the first iteration, the deviance is 802.17. This is close to the final GLM deviance of 716.85, but the model has not converged yet.

```
w1 = mu1
z1 = log(mu1) + (y - mu1) / mu1
fit2 = lm(z1 ~ Area + Elevation + Nearest + Scruz + Adjacent,
          weights = w1, data = gala)

coef(fit2)
```

```
##   (Intercept)          Area     Elevation       Nearest         Scruz
##  3.1646548788 -0.0005731435  0.0035188301  0.0077916350 -0.0050690616
##      Adjacent
## -0.0006714962
```

```
eta2 = fitted(fit2)
mu2 = exp(eta2)

dev2 = 2 * sum(y * log(y / mu2) - (y - mu2))
dev2
```

```
## [1] 718.5076
```

After the second iteration, the coefficients are 3.16465, -0.0005731, 0.0035188, 0.0077916, -0.0050691, and -0.0006715. The deviance is 718.51, which is close to the final GLM deviance of 716.85, this could mean its starting to converge.

```
w2 = mu2
z2 = log(mu2) + (y - mu2) / mu2

fit3 = lm(z2 ~ Area + Elevation + Nearest + Scruz + Adjacent,
          weights = w2, data = gala)

coef(fit3)
```

```
##   (Intercept)          Area     Elevation       Nearest         Scruz
##  3.1548680107 -0.0005796951  0.0035396925  0.0087886455 -0.0056816338
##      Adjacent
## -0.0006633494
```

```
eta3 = fitted(fit3)
mu3 = exp(eta3)

dev3 = 2 * sum(y * log(y / mu3) - (y - mu3))
dev3
```

```
## [1] 716.8482
```

```
w3 = mu3
z3 = log(mu3) + (y - mu3) / mu3

fit4 = lm(z3 ~ Area + Elevation + Nearest + Scruz + Adjacent,
          weights = w3, data = gala)

coef(fit4)
```

```
##   (Intercept)          Area     Elevation       Nearest         Scruz
##  3.1548078334 -0.0005799425  0.0035405926  0.0088255110 -0.0057093743
##      Adjacent
## -0.0006630316
```

```
eta4 = fitted(fit4)
mu4 = exp(eta4)

dev4 = 2 * sum(y * log(y / mu4) - (y - mu4))
dev4
```

```
## [1] 716.8458
```

After the fourth iteration, the deviance is 716.8458 and is no longer changing much. The final estimated coefficients are also nearly identical to those from the direct GLM fit. Therefore, its converged.

```
X = model.matrix(fit4)
W = diag(w3)

se_manual = sqrt(diag(solve(t(X) %*% W %*% X)))

se_manual
```

```
##  (Intercept)         Area    Elevation      Nearest        Scruz     Adjacent
## 5.173731e-02 2.626728e-05 8.738306e-05 1.820740e-03 6.247521e-04 2.932464e-05
```

The SE from the final iterated linear model are 0.05174, 0.00002627, 0.00008738, 0.001821, 0.0006248, and 0.00002932. These are almost identical to the SE from the direct GLM fit.